COUNTEREXAMPLES TO THE ZHANG–WAN DEEP-HOLE COMPLETENESS CONJECTURE

Abstract. Let $E/\mathbb{F}_q$ be an elliptic curve of odd characteristic, let $N = \#E(\mathbb{F}_q) \geq q+4$, and put $D = E(\mathbb{F}_q) \setminus \{O\}$. Zhang and Wan conjectured that, for $2 \leq k \leq N-4$, the layer $C_{\mathcal{L}}(D, (k+1)O) \setminus C_{\mathcal{L}}(D, kO)$ is the complete set of deep holes of $C_{\mathcal{L}}(D, kO)$. We refute the conjecture as stated. At the upper endpoint $k = N-4$, the code has covering radius two and $(q-1)(q^2+q+2-N)$ deep-hole cosets, while the proposed layer contains only $q-1$ cosets. We also give an example over $\mathbb{F}_3$ for which the conjecture fails at both admissible values $k=2$ and $k=3$.

For a linear code $C \subseteq \mathbb{F}_q^n$, let $\rho(C)$ denote its covering radius. A word $v \in \mathbb{F}_q^n$ is a *deep hole* if $d(v, C) = \rho(C)$. Distance to C is constant on cosets of C.

Zhang and Wan [2, Conjecture 1.4] conjectured the following. Let $E/\mathbb{F}_q$ be an elliptic curve, where q is odd, with $N = \#E(\mathbb{F}_q) \geq q+4$. For $O \in E(\mathbb{F}_q)$ and $D = E(\mathbb{F}_q) \setminus \{O\}$, they asserted that

$$C_{\mathcal{L}}(D, (k+1)O) \setminus C_{\mathcal{L}}(D, kO) \tag{1}$$

is the set of all deep holes of $C_{\mathcal{L}}(D, kO)$ whenever $2 \leq k \leq N-4$. We first show that the upper endpoint always fails.

1. The endpoint obstruction

Lemma 1. *Let C be an $[n, n-3]$ code over $\mathbb{F}_q$ with $d(C) \geq 3$. If*

$$q+1 < n < q^2+q+1,$$

then $\rho(C) = 2$, and the number of deep-hole cosets of C is

$$q^3 - 1 - n(q-1) = (q-1)(q^2+q+1-n).$$

Proof. Let H be a $3 \times n$ parity-check matrix. Since $d(C) \geq 3$, the columns of H are nonzero and pairwise nonproportional, and hence define n distinct points of $\mathrm{PG}(2, q)$.

Let $s \in \mathbb{F}_q^3 \setminus \{0\}$. If $[s]$ is a column point, then s is generated by one column. Otherwise, the $q+1$ projective lines through $[s]$ partition the column points. Since $n > q+1$, one of these lines contains two column points, whose representatives span s. Thus every syndrome is generated by at most two columns, so $\rho(C) \leq 2$.

Since $\mathrm{PG}(2, q)$ has q^2+q+1 points and $n < q^2+q+1$, some nonzero syndrome is not proportional to a column. Its coset weight is two, and

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therefore $\rho(C) = 2$. Of the q^3 syndromes, one has weight zero and exactly $n(q-1)$ have weight one. Every remaining syndrome has weight two, giving the stated count. $\square$

Theorem 2. *Let q be an odd prime power, let $E/\mathbb{F}_q$ be an elliptic curve with $N = \#E(\mathbb{F}_q) \geq q + 4$, and fix $O \in E(\mathbb{F}_q)$. Put*

$$D = E(\mathbb{F}_q) \setminus \{O\}, \qquad n = N - 1, \qquad k = N - 4 = n - 3,$$

and let $C = C_{\mathcal{L}}(D, kO)$. Then $\rho(C) = 2$, and C has

$$(q-1)(q^2 + q + 1 - n)$$

deep-hole cosets. The layer $C_{\mathcal{L}}(D, (k+1)O) \setminus C$ is the union of exactly $q-1$ deep-hole cosets and is a proper subset of the deep holes of C. Consequently, Conjecture 1.4 of Zhang and Wan is false as stated; its hypotheses are satisfiable, as Proposition 3 shows.

Proof. Since E has genus one, Riemann–Roch gives $\ell(kO) = k$ for $k \geq 1$ (here $k = N - 4 \geq q \geq 3$). Evaluation on D is injective because $\deg(kO - D) = k - n < 0$. Hence C has dimension $k = n - 3$, and the standard designed-distance bound gives $d(C) \geq n - k = 3$; see [1, Chapter 2].

The hypothesis $N \geq q + 4$ and Hasse's bound give

$$q + 3 \leq n \leq q + 2\sqrt{q}.$$

Thus $q + 1 < n < q^2 + q + 1$, and Lemma 1 gives the covering radius and the number of deep-hole cosets.

Set $C' = C_{\mathcal{L}}(D, (k+1)O)$. Again by Riemann–Roch and injectivity of evaluation,

$$\dim C' = k + 1, \qquad d(C') \geq n - (k+1) = 2.$$

Thus C'/C is one-dimensional. Each of its $q-1$ nonzero cosets has coset weight at least two, since C' contains no word of weight one. As $\rho(C) = 2$, these are deep-hole cosets.

Finally,

$$n \leq q + 2\sqrt{q} < q^2 + q,$$

so $q^2 + q + 1 - n > 1$. Hence the total number of deep-hole cosets is strictly larger than $q - 1$, and the layer is not complete. $\square$

Two of the assertions of Theorem 2 are already those of the source. Since $\#E(\mathbb{F}_q) - n = 1$ for $D = E(\mathbb{F}_q) \setminus \{O\}$, the count $(\#E(\mathbb{F}_q) - n)(q-1)q^k$ of [2, Theorem 1.2(iii)–(iv)] specialises to $(q-1)q^k$ words, that is, to the $q-1$ cosets of the layer, and the hypothesis $k < n - 2$ of their part (iv) holds at $k = n - 3$; likewise $\rho(C) = n - k - 1 = 2$ is their part (ii). What Theorem 2 adds is, first, that both hold with no side condition: their theorem requires, besides $n \geq q+3$, one of $n \geq q+k$, q prime, or $k \leq \sqrt{q}$, and at $k = n-3$ the first forces $q \leq 3$ while the third fails because $k \geq q$, so for non-prime q their argument does not reach the endpoint, whereas Lemma 1 gives $\rho(C) = 2$ unconditionally. Second, and this is the point, it supplies the total number

$(q-1)(q^2+q+1-n)$ of deep-hole cosets, against which the layer can be compared.

The cap $k \le N-4$ is deliberate: Zhang and Wan note in their Remark 1.3 that at $k = n-2 = N-3$ the covering radius drops to $n-k-1 = 1$ (again under one of those conditions), so every word outside C is a deep hole and completeness fails there for a trivial reason. Theorem 2 shows that it already fails at $k = N-4$, the largest value the conjecture does admit, and Proposition 3 shows it can fail at the smallest admissible value $k = 2$ as well; the failure is therefore not an artefact of the endpoint. We note finally that [2, Corollary 4.11] does prove a conditional completeness statement in codimension three, but for the residue codes $C_\Omega(D, kO)$ rather than for the functional codes $C_{\mathcal{L}}(D, kO)$ of Conjecture 1.4; it does not conflict with Theorem 2.

2. Failure at both parameters over $\mathbb{F}_3$

Proposition 3. *Over $\mathbb{F}_3$, let*

$$E : \; y^2 = x^3 + 2x + 1,$$

let O be the point at infinity, and put $D = E(\mathbb{F}_3) \setminus \{O\}$. Then $\#E(\mathbb{F}_3) = 7$, and the Zhang–Wan conjecture fails for both admissible parameters $k = 2$ and $k = 3$.

Proof. The projective closure is

$$F(X,Y,Z) = Y^2Z - X^3 - 2XZ^2 - Z^3 = 0.$$

Over $\mathbb{F}_3$, one has $F_X = Z^2$, so a singular point would have $Z = 0$. The unique point at infinity is $O = [0 : 1 : 0]$, and $F_Z(O) = 1$; hence E is nonsingular. Moreover, $x^3 + 2x + 1 = 1$ for every $x \in \mathbb{F}_3$, so $\#E(\mathbb{F}_3) = 7$. Order the points of D as

$$(0,1),\ (0,2),\ (1,1),\ (1,2),\ (2,1),\ (2,2);$$

all coordinates below refer to this order. The functions x and y have their only poles at O, of orders two and three, respectively. Hence Riemann–Roch gives

$$L(2O) = \langle 1, x\rangle, \qquad L(3O) = \langle 1, x, y\rangle.$$

First take $k = 2$ and write $C_2 = C_{\mathcal{L}}(D, 2O)$. Then

$$\begin{aligned} C_2 &= \{(a, a, a+b, a+b, a+2b, a+2b) : a, b \in \mathbb{F}_3\} \\ &= \{(u_0, u_0, u_1, u_1, u_2, u_2) : u_0 + u_1 + u_2 = 0\}. \end{aligned}$$

View a received word as three consecutive coordinate pairs. Every word of $\mathbb{F}_3^6$ agrees with a word of C_2 in at least three coordinates. Indeed, if one pair is constant, match both entries there, match one entry in a second pair, and choose the value on the third pair to make $u_0 + u_1 + u_2 = 0$. If all three pairs are nonconstant, let $A_i \subset \mathbb{F}_3$ be the two symbols occurring in the ith pair. Choose $u_2 \in A_2$. Since any two two-element subsets of $\mathbb{F}_3$

satisfy $A_0 + A_1 = \mathbb{F}_3$, there are $u_0 \in A_0$ and $u_1 \in A_1$ with $u_0 + u_1 = -u_2$, matching one entry in every pair. Thus $\rho(C_2) \leq 3$.

Now let

$$w = (0, 1, 0, 1, 0, 2).$$

Every pair of w is nonconstant, so every word of C_2 differs from w in at least one coordinate of each pair; hence $d(w, C_2) \geq 3$. Since $\mathrm{wt}(w) = 3$ and $0 \in C_2$, this lower bound is attained, so $d(w, C_2) = 3$. Hence $\rho(C_2) = 3$ and w is a deep hole. For $f = a+bx+cy \in L(3O)$, the difference $f(x, 1)-f(x, 2) = -c$ is independent of x. The three corresponding pair differences of w are $2, 2, 1$, so $w \notin C_{\mathcal{L}}(D, 3O)$. Therefore w is a deep hole of C_2 outside the proposed layer $C_{\mathcal{L}}(D, 3O) \setminus C_2$. Since $\dim C_{\mathcal{L}}(D, 3O) = 3$, that layer is the union of exactly two cosets of C_2, and both are deep-hole cosets: every nonzero word of $C_{\mathcal{L}}(D, 3O)$ has weight at least $6 - 3 = 3$, so each of the two has coset weight $3 = \rho(C_2)$. Thus at $k = 2$ the layer is made of deep holes and is still incomplete, w exhibiting a third deep-hole coset. A direct enumeration of $\mathbb{F}_3^6$, on which nothing here depends, shows that 24 of the 81 cosets of C_2 have coset weight three, so that 22 of them lie outside the layer.

For $k = 3 = N - 4$, Theorem 2 applies. The code $C_{\mathcal{L}}(D, 3O)$ has fourteen deep-hole cosets, whereas $C_{\mathcal{L}}(D, 4O) \setminus C_{\mathcal{L}}(D, 3O)$ consists of only two cosets. Thus the conjecture also fails at $k = 3$. $\square$

Exact verification. The one finite computation quoted above is the coset census at the end of the proof of Proposition 3, and it is recomputable from the data printed in this paper. From $L(2O) = \langle 1, x\rangle$ and the point order fixed in that proof one writes down the nine words of C_2, forms the 81 cosets of C_2 in $\mathbb{F}_3^6$, and takes the minimum weight in each: exactly 24 have coset weight three, and since the layer $C_{\mathcal{L}}(D, 3O) \setminus C_2$ accounts for two of these, 22 of them lie outside the layer. The same printed data carry the two checks the argument actually uses, both of them by hand: $d(w, C_2) = 3$ for the word w displayed above, and $w \notin C_{\mathcal{L}}(D, 3O)$ because the three pair differences of w are not constant. The count of fourteen deep-hole cosets of $C_{\mathcal{L}}(D, 3O)$ is Lemma 1 evaluated at $q = 3$ and $n = 6$, not a computation. All arithmetic is exact, in $\mathbb{F}_3$ and on integers; no floating-point computation is involved.

The census was also re-run independently, by a program written from the definitions of this note rather than from the first implementation: its 81 coset-weight determinations, and the two counts 24 and 22 derived from them, agreed throughout with the values printed above. Both programs and the transcript of that run are retained in an auxiliary archive and are available on request; the specification above is what a reader should reimplement. A program accompanying this note carries out that reimplementation: from the field, the curve, the point order and the word printed above it re-derives C_2, the 81 coset weights and the counts 24 and 22, and also the covering radius and the deep-hole count of Theorem 2 for this curve and for a bounded family of curves over small prime fields. That archive is separate from it and is not distributed with this note. No part of Theorem 2 or of Proposition 3 depends on the census, which is reported only to quantify how far the layer falls short at $k = 2$.

Status and scope

Theorem 2 refutes Conjecture 1.4 of [2] as stated: for every odd prime power q, every elliptic curve $E/\mathbb{F}_q$ with $\#E(\mathbb{F}_q) \geq q+4$ and every $O \in E(\mathbb{F}_q)$, the conjecture fails at $k = N - 4$; Proposition 3 exhibits an instance in which it also fails at $k = 2$. What is not settled here is the interior of the range. For $q > 3$ we do not decide, for any single $k < N - 4$, whether $C_{\mathcal{L}}(D, (k+1)O) \setminus C_{\mathcal{L}}(D, kO)$ is the complete set of deep holes of $C_{\mathcal{L}}(D, kO)$, and we do not propose a corrected range. In both cases treated here the proposed layer does consist of deep holes, so what fails is completeness alone.

The conjecture is quoted from the preprint version of [2], whose statement numbering we follow; we were not able to consult the printed version, in which the numbering may differ. We are aware of no resolution of Conjecture 1.4 in the literature and of no earlier appearance of the word w of Proposition 3; that is a bounded negative obtained from the citing literature and not a proof of absence. No novelty is claimed for Lemma 1, which is a standard syndrome count in $\mathrm{PG}(2, q)$; the same geometric viewpoint is used in [2, Proposition 4.9].

References

[1] H. Stichtenoth, *Algebraic Function Fields and Codes*, 2nd ed., Graduate Texts in Mathematics, vol. 254, Springer, Berlin, 2009.

[2] J. Zhang and D. Wan, *On deep holes of elliptic curve codes*, IEEE Trans. Inform. Theory **69** (2023), no. 7, 4498–4506, doi:10.1109/TIT.2023.3257320; preprint arXiv:2207.12584. Statement numbers cited above are those of the preprint, in which the completeness conjecture is Conjecture 1.4, the covering-radius and deep-hole theorem is Theorem 1.2, the boundary case $k = n - 2$ is discussed in Remark 1.3, and the conditional completeness result for residue codes is Corollary 4.11.